Certified sign classification of the second variation of the harmonic zero-point energy for Gaussian interactions on $a \in [0.425, 4]$: not a local minimum for $a \in [0.425, 1.579]$ and a strict minimum along the shear for $a \in [1.584, 4]$

(a partial result: the full "iff" for all $a > 0$ is not claimed)

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Abstract

Let $E(a,t)$ be the harmonic zero-point energy per particle ($\hbar = m = 1$) of the planar Bravais lattices $L_t = P_t L_\rho$, $P_t = \left(\begin{smallmatrix} 1 & t \\ t & 1+t^2 \end{smallmatrix}\right)$, sheared from the triangular lattice L_ρ, for the Gaussian pair potential $V(r) = e^{-\pi a r^2}$, and let $F(a,t) = \partial_t^2 E(a,t)$ be its *exact* second variation along the shear (not the majorant of the predecessor B2, which cannot reach the sign changes). We enclose F two-sided by ball arithmetic, for *every* a in an interval and every $|t| \le 10^{-8}$, and chain 174 such intervals. **Result.** $F < 0$ on all of [0.425, 1.579] and $F > 0$ on all of [1.584, 4]. Hence along the shear the triangular lattice is *not a local minimum* of the zero-point energy for every $a \in [0.425, 1.579]$ ($E(a,t) < E(a,0)$ for $0 < |t| \le 10^{-8}$), and $E(a,0) < E(a,t)$ for every $a \in [1.584, 4]$ (a *strict minimum along the shear*; a strict local minimum over lattice shapes needs further assumptions, stated). Each of $(0.3745, 0.379)$ and $(1.579, 1.583)$ contains a zero of $a \mapsto F(a,0)$ (intermediate value theorem); write a_- and a_+ for such zeros. Uniqueness, and the absence of other zeros in the uncertified gaps, are not certified. **Not claimed.** The full equivalence "local minimum $\iff$ $a > a_+$" for all $a > 0$ is *not* claimed and is probably false: the earlier preprint P2 sees another sign change near $a \approx 0.123$ (Tier [X]). On the left of a_- only isolated points are certified ($F > 0$ at $a = 0.33, 0.35, 0.36, 0.37, 0.372, 0.3745$); the intervals in $(0.383, 0.425)$ and everything below 0.33 are not certified (at $a = 0.25$ and 0.30 the certifier could not prove $\det B > 0$). **Tier split.** [A] (Lean 4.34.1 + Mathlib, axioms within {propext, Classical.choice, Quot.sound}): the logic, the signs of 197 rational records, the interval-chain coverage, the intermediate-value step and the window lemmas; the theorems take the enclosures as hypotheses. [C-ball]: the Arb enclosures (1,276,022 certified cells over 202 distinct run records, 0 failed), not kernel-checked. [C-human]: that F is the second derivative of E (so E is C^2 in t), the mirror symmetry, continuity in a, and, for the strict-minimum claim over shapes, criticality, isotropy and the second-derivative test. The headline is therefore Tier C overall. Controls: just inside ($a = 0.40$), a tampered certificate, straddling intervals and a planted false LeanFlow sign are all rejected; a 609-check floating-point cross-check (3 per run record, duplicate included) has 0 misses. The kernel replay (leanchecker) of `Cert`, `Controls`, `Honest` and `Basic` passes with rc 0.

1 Setting, predecessors, and what is and is not claimed

Predecessors. P2 (doi:10.5281/zenodo.23653833, [X]) conjectured (QU-1) that the triangular lattice minimises E_{zp} among unit-covolume lattices for completely monotone potentials and refuted it numerically for the

Gaussian $e^{-\pi a r^2}$ at some a: P2's curvature $c(a) = \operatorname{Lap} E_{\mathrm{zp}}/4$ changes sign near $a \approx 0.3767$ and $a \approx 1.5815$ and, in addition, near $a \approx 0.123$ (it is negative at $a = 0.1$), and its values at $a = 6, 8$ are flagged for precision. B2 (doi:10.5281/zenodo.23655392) certified, at the single value $a = 1$, that ρ is not a local minimum, via a linear-response majorant of E. N2 (doi:10.5281/zenodo.23663551) certified the Yukawa case via a moving comparison matrix with exact curvature. N1, this paper, goes back to the Gaussian family and asks for the *sign pattern in a*, which needs the exact second variation rather than a majorant (Step 1 below).

Tiers. [A]: kernel-checked Lean (pinned Lean 4.34.1 and Mathlib; every declaration prints axioms within propext, Classical.choice, Quot.sound; `sorryAx` and smuggled axioms rejected). [L]: literature or own earlier preprints, cited. [C-ball]: rigorous Arb ball arithmetic (python-flint 0.6.0, 96 bits, outward rounded), not kernel-checked. [C-human]: a written argument, checked by no machine and no human. [X]: floating point only, never supporting a claim. A claim is filed no higher than what it rests on.

What is claimed. (i) $F < 0$ on $[0.425, 1.579]$ and $F > 0$ on $[1.584, 4]$, for all $|t| \le 10^{-8}$ [A]+[C-ball]; (ii) the two root enclosures [A]+[C-ball]+[C-human]; (iii) the consequences along the shear path [A]+[C-ball]+[C-human].
What is not claimed. (a) The iff for all $a > 0$. (b) Anything for $a < 0.33$ or $a > 4$. (c) Uniqueness of the roots. (d) A strict local minimum over shapes without the extra assumptions of Section 6. (e) Any statement about the total (classical plus zero-point) energy, other lattices, other potentials, or three dimensions. The window $|t| \le 10^{-8}$ is very small; the statements are local to the shear path.

2 Results at a glance

Notation. $E(a,t)$ and $F(a,t) = \partial_t^2 E(a,t)$ as above; $t_0 = 10^{-8}$; $c(a) = F(a,0)/8 = \operatorname{Lap} E_{\mathrm{zp}}/4$ (the curvature of P2; $F(a,0) = 2 \operatorname{Lap} E_{\mathrm{zp}}(\rho)$ under the hypotheses of Section 6, [C-human]). All "Lean" names are in module `ProgZeroPointClass` (files `Basic.lean`, `Cert.lean`; listings transliterated to ASCII; full text in `data/lean/`).

Statement	Tier	Lean / evidence
T1. $F(a,t) < 0$ for all $a \in [0.425, 1.579]$, $\lvert t\rvert \le t_0$	[A] logic; [C-ball] enclosures (71 records)	`segment_neg0`, `chain_neg0`, `sign_neg`; ledger `PZC-C-0002`
T2. $F(a,t) > 0$ for all $a \in [1.584, 4]$, $\lvert t\rvert \le t_0$	[A]; [C-ball] (103 records)	`segment_pos0`, `chain_pos0`, `sign_pos`; `PZC-C-0001`
T3. $a \mapsto F(a,0)$ has a zero in $(0.3745, 0.379)$ and in $(1.579, 1.583)$	[A] (IVT); [C-ball] (4 thin points); [C-human] (continuity in a)	`classification`, `root_in_gap`; `PZC-C-0003`
T4. Inside: ρ not a local minimum along the shear, $E(a,t) < E(a,0)$ for $0 < \lvert t\rvert \le t_0$	as T1; [C-human] (C^2, mirror symmetry)	`segment_neg0_path`
T5. Outside: $E(a,0) < E(a,t)$ for $0 < \lvert t\rvert \le t_0$ (strict minimum along the shear)	as T2; [C-human]	`segment_pos0_path`
T6. Outside: strict local min over unit-covolume lattice shapes	T5 plus [C-human] (criticality, isotropy, 2D test), not formalised	Section 6
T7. Every zero of $a \mapsto F(a,t)$ in $[0.425, 4]$ lies in $(1.579, 1.584) \setminus \{1.583\}$	[A] logic (not formalised as one theorem); [C-ball] (T1, T2 and the 1.583 record)	no separate theorem

Table 1: The theorems of this paper and their tiers. T1–T7 are about the Gaussian family on the stated ranges only.

3 Step-by-step human demonstration, with the matching Lean statements

Step 1: why B2's majorant cannot work near the roots [X]+[C-human]. B2 bounds E above by a fixed-matrix majorant U and proves $E''(0) \le U''(0) < 0$ when $U''(0) < 0$. The gap is a relaxation term: $U''(0) - E''(0) = \frac{1}{2}\int \mathrm{tr}(X^{-1}Y^2)\, d\kappa \ge 0$ (N2, `PZP-C-0008`, [C-human]), where $X = D_0^{1/2}$ and $XY + YX = D_0'$. So $U'' \ge E''$ always. At a root of $F(\cdot, 0)$ one has $E'' = 0$, hence $U'' \ge 0$ there, and a negative U'' is impossible. Numerically ([X], $N = 64$, `runs/explore_tierX_N64.txt`) it is worse than the inequality demands: $U'' = +2.79$ at the root 0.3767 and $+8.77$ at the root 1.5815 (both > 0), and $U''(0) < 0$ only at $a = 0.5, 0.7, 1$ of the 17 grid values (-1.013 at 0.5; $+3.877$ at 1.3), so its sign changes lie in $(0.4, 0.5)$ and $(1.0, 1.3)$. No majorant of this type can classify the signs near $a_\pm$, and none gives a positive lower bound at all (it is an upper bound). We therefore enclose F itself.

Step 2: the exact second variation [C-human] (`PZC-C-0009`). Let $H_n(t) = \mathrm{Hess}\, e^{-\pi a |P_t R_n|^2}$, $D_t(\kappa) = \sum_{n\neq 0}(1 - \cos 2\pi\kappa\cdot n) H_n(t)$ and $E(a,t) = \frac{1}{2}\int \mathrm{tr}\, D_t^{1/2}\, d\kappa$ over the Brillouin zone (B2's setting). The matrix D_t is 2×2, symmetric, and positive definite for $\kappa \neq 0$ when the lattice is dynamically stable. *Identity.* For $S = D^{1/2}$, Cayley–Hamilton gives $(\mathrm{tr}\, S)^2 = \mathrm{tr}\, S^2 + 2\det S = \mathrm{tr}\, D + 2\sqrt{\det D}$, hence

$$\mathrm{tr}\sqrt{D} = \sqrt{\mathrm{tr}\, D + 2\sqrt{\det D}}.$$

With the Duffy coordinates of B2 ($\kappa = s\, v(t')$, Jacobian $s/2$, $D_t = s^2 B_t$, $B_t = \sum_n \theta_n^2 f(s\theta_n) H_n(t)$, $f(x) = (1-\cos x)/x^2$) we get $\mathrm{tr}\sqrt{D_t} = s\,\tau$ with $\tau = \sqrt{T + 2\delta}$, $T = \mathrm{tr}\, B_t$, $\delta = \sqrt{\det B_t}$, and $F = \int g$ with the regular integrand $g = s^2\tau''/4$ (the prefactors and triangle bookkeeping are those of B2 and N2, and so is the Brillouin-zone coverage; both are inherited, filed [C-human], and checked here only at Tier [X] by agreement with P2's independent $c(a)$ and roots. A positive overall prefactor cannot change any sign claimed here, whereas a missing Brillouin-zone region could). Differentiating twice in t gives closed forms:

$$\delta'' = \frac{(\det B)''}{2\delta} - \frac{((\det B)')^2}{4\delta^3}, \qquad \tau'' = \frac{T'' + 2\delta''}{2\tau} - \frac{(T' + 2\delta')^2}{4\tau^3},$$

with B', B'' from the closed forms of H_n', H_n'' (context only: B2's Lean `hasDerivAt_*` cover the scalar pieces, but no N1 theorem uses them). There is no inequality in this step: F is the exact second derivative, and the only singularities are where $\det B = 0$ or $\mathrm{tr}\, B = 0$, which Step 3 excludes. [X] check of the closed form against finite differences of E (17 values of a): agreement to $\le 4.92\times 10^{-4}$ (at $a = 2$), e.g. $E''(0) = -7.695875$ (closed form) against -7.695997 (FD) at $a = 1$, and $c = E''/8$ reproduces P2's $c(a)$ (0.18806 against 0.188 at $a = 0.25$). Lean: the identity and the closed forms are *not* formalised; Lean sees F only through its enclosure and the hypothesis `hE'` of Step 6.

Step 3: an enclosure affine in a, which allows chaining [C-ball] (`PZC-C-0001..0004`). For an interval $[a_c - \Delta, a_c + \Delta]$ (a thin point has $\Delta = 0$) and the window $|t| \le t_0$ (entering as the ball $[-t_0, t_0]$ in $H_n(t)$), the certifier `n1_certify2.py` returns three balls Q, D, R with

$$F(a,t) - (a - a_c)\, d \;\in\; Q + R \quad \text{for some } d \in D, \text{ for every } a \text{ in the interval and every } |t| \le t_0.$$

Q and D are midpoint cell sums of g and $\partial_a g$ at the exact centre a_c; R is the second-order Lagrange remainder in (s, t', a) on cell $\times$ interval (first-order terms in s, t' integrate to zero on a square cell, the first-order term in a is $(a - a_c)d$). Why affine: a plain ball in a of half-width Δ would cost a first-order error $\Delta \sup|\partial_a F|$, which swamps F near the roots where $F \approx 0$; the affine form leaves only the second-order remainder $O(\Delta^2)$ in R, so the sign certificate $q_{hi} + c_{\max} < 0$ (resp. $q_{lo} + c_{\min} > 0$) survives for intervals of width 0.001 to 0.054. Here $c_{\min}, c_{\max}$ are the extreme corner values of $(a - a_c)d$ over $[a_{lo}, a_{hi}] \times [d_{lo}, d_{hi}]$. A cell counts only if $\det B > 0$ and $\mathrm{tr}\, B > 0$ are certified on the whole cell ball $\times$ a-interval $\times$ t-window; a failed cell makes the record uncertified and it never enters Lean (Section 4). In Lean the record and its meaning are:

```
structure Rec where  lo hi ac qlo qhi dlo dhi : Q
def Rec.cmax r := max (max ((lo-ac)*dlo) ((lo-ac)*dhi)) (max ((hi-ac)*dlo) ((hi-ac)*dhi))   -- cmin: min
def allNeg (L : List Rec) : Bool := L.all (fun r => decide (r.qhi + r.cmax < 0))
```

```
def allPos (L : List Rec) : Bool := L.all (fun r => decide (0 < r.qlo + r.cmin))
def Encloses (F : R -> R -> R) (t0 : R) (L : List Rec) : Prop :=
  forall r in L, forall a, r.lo <= a -> a <= r.hi -> forall t, |t| <= t0 ->
    exists d, r.dlo <= d /\ d <= r.dhi /\ r.qlo <= F a t - (a - r.ac)*d /\ F a t - (a - r.ac)*d <= r.qhi
```

`Encloses` *is exactly the [C-ball] input*: it is a hypothesis of every assembled theorem, never proved. The lemmas `mul_le_corners` and `corners_le_mul` ([A]) bound $(a - a_c)d$ by the corner values; `rec_upper` and `rec_lower` ([A]) turn an `Encloses` record into $F \le q_{hi} + c_{\max}$ and $F \ge q_{lo} + c_{\min}$.

Step 4: coverage of $[A, B]$ by a chain of intervals [A] (`PZC-A-0002, -0003`). A list of records chains from A to B if the first starts at A, consecutive records satisfy $hi_i = lo_{i+1}$, and the last ends at B. Then every real $a \in [A, B]$ lies in some record. The kernel evaluates the Boolean on the actual lists by `decide +kernel`:

```
def chainFrom : Q -> Q -> List Rec -> Bool
  | _, _, [] => false
  | x, B, r :: rs => decide (r.lo = x) && decide (r.lo <= r.hi) && (decide (r.hi = B) || chainFrom r.hi B rs)
theorem covers_of_chain : forall L A B, chainFrom A B L = true ->
    forall a : R, A <= a -> a <= B -> exists r in L, r.lo <= a /\ a <= r.hi
theorem chain_neg0 : chainFrom (17/40) (1579/1000) Lnegseg0 = true := by decide +kernel
theorem chain_pos0 : chainFrom (198/125) 4 Lposseg0 = true := by decide +kernel
theorem sign_neg : allNeg Lneg = true := by decide +kernel
theorem sign_pos : allPos Lpos = true := by decide +kernel
```

`Lnegseg0` has 71 records from $17/40 = 0.425$ to $1579/1000$, `Lposseg0` has 103 from $198/125 = 1.584$ to 4; `Lneg` and `Lpos` add the thin points and one more interval (197 records in all, 113 positive and 84 negative). The kernel checks that the records chain (no gap, no overlap beyond endpoints) and that every one has its sign. Assembling with Step 3:

```
theorem segment_neg0 (F : R -> R -> R) (h : Encloses F t0 Lneg) :
    forall a, 17/40 <= a -> a <= 1579/1000 -> forall t, |t| <= t0 -> F a t < 0
theorem segment_pos0 (F : R -> R -> R) (h : Encloses F t0 Lpos) :
    forall a, 198/125 <= a -> a <= 4 -> forall t, |t| <= t0 -> 0 < F a t
```

This is T1 and T2, in the exact form: conditional on `Encloses`.

Step 5: the roots by the intermediate value theorem [A]+[C-human] (`PZC-A-0005, PZC-C-0011`). The thin-point records at $a = 0.3745$ ($F \in [0.0177, 0.0949]$) and $a = 0.379$ ($F \in [-0.0978, -0.0235]$) have opposite certified signs, and so do $a = 1.579$ ($F \in [-0.0423, -0.0195]$) and $a = 1.583$ ($F \in [0.0075, 0.0303]$) (values at $t \in [-t_0, t_0]$). If $a \mapsto F(a, 0)$ is continuous on each bracket ([C-human]: it is analytic in a, being a uniformly convergent lattice sum of smooth integrals; not formalised), the Mathlib intermediate value theorem gives a zero strictly inside:

```
theorem root_in_gap (f : R -> R) (hlt : aL < aR) (hf : ContinuousOn f (Icc aL aR)) (hL : 0 < f aL) (hR : f aR
    < 0) :
    exists a in Ioo aL aR, f a = 0                      -- root_in_gap' : the mirrored signs
theorem classification (F) (hp : Encloses F t0 Lpos) (hn : Encloses F t0 Lneg)
    (hcm : ContinuousOn (fun a => F a 0) (Icc 0.3745 0.379)) (hcp : ContinuousOn (fun a => F a 0) (Icc 1.579
    1.583)) :
    (forall r in Lpos, ... 0 < F a t) /\ (forall r in Lneg, ... F a t < 0) /\
    (exists a in Ioo 0.3745 0.379, F a 0 = 0) /\ (exists a in Ioo 1.579 1.583, F a 0 = 0)
```

(`brackets_ordered`, a `decide +kernel` fact, checks that the bracket points are the thin records with $lo = hi$ and are ordered.) P2's [X] roots 0.37668 and 1.58148 lie inside the two intervals. Uniqueness is *not* certified: no sign of $\partial_a F$ was certified on the enclosures. T7 follows from T1, T2 and the thin record at 1.583 (so it rests on the [C-ball] enclosures; no Lean theorem states it): a zero of $F(\cdot, t)$ in $[0.425, 4]$ cannot lie in $[0.425, 1.579]$ or $[1.584, 4]$, and not at 1.583.

Step 6: from the sign of F to the energy along the path [A] logic, [C-human] inputs (`PZC-A-0001, -0006`). *Window lemma.* If E is differentiable on $[-t_0, t_0]$ with derivative E', E' is differentiable with derivative F, and $F \le B$ there, then $E(t) - \frac{B}{2}t^2$ has second derivative ≤ 0, hence is concave, and

$$E(t) + E(-t) - 2E(0) \le B\,t^2 \qquad (|t| \le t_0);$$

with $F \geq B$ the function is convex and the inequality reverses. (Mathlib `concaveOn_of_hasDerivWithinAt2_nonpos` and the convex twin.) With the mirror symmetry $E(-t) = E(t)$ this gives, for $B < 0$: $E(t) < E(0)$ for $0 < |t| \leq t_0$, so 0 is not a local minimum; for $B > 0$: $E(0) < E(t)$.

```
theorem window_upper {E E' E'' : R -> R} {t0 B : R} (hE : forall t in Icc (-t0) t0, HasDerivAt E (E' t) t)
    (hE' : forall t in Icc (-t0) t0, HasDerivAt E' (E'' t) t) (hB : forall t in Icc (-t0) t0, E'' t <= B)
    (ht : t in Icc (-t0) t0) : E t + E (-t) - 2 * E 0 <= B * t ^ 2      -- window_lower : >=
theorem not_localMin_of_curv_neg (ht0 : 0 < t0) (hBneg : B < 0) (hE) (hE') (hB) (hsym : forall t, E (-t) = E t
    ) :
    Not (IsLocalMin E 0) /\ forall t, 0 < t -> t <= t0 -> E t < E 0
theorem strict_gt_of_curv_pos (hBpos : 0 < B) (hE) (hE') (hB) (hsym) :
    forall t, t != 0 -> |t| <= t0 -> E 0 < E t
theorem segment_neg0_path (E E' F) (h : Encloses F t0 Lneg) (hE : forall a t, t in Icc (-t0) t0 -> HasDerivAt
    (E a) (E' a t) t)
    (hE' : forall a t, ... -> HasDerivAt (E' a) (F a t) t) (hsym : forall a t, E a (-t) = E a t) :
    forall a, 17/40 <= a -> a <= 1579/1000 -> Not (IsLocalMin (E a) 0) /\ forall t, 0 < t -> t <= t0 -> E a t
    < E a 0
```

`segment_pos0_path` is the same with $[198/125, 4]$ and conclusion $E\,a\,0 < E\,a\,t$ for $t \neq 0$; `along_path` gives the per-record version, which includes the thin points. The hypotheses `hE`, `hE'`, `hsym` are exactly the [C-human] inputs: (1) $E(a, \cdot)$ is C^2 with second derivative F on the window (differentiation under the Brillouin-zone integral: the Duffy integrand is jointly smooth and bounded on the compact domain because $\det B > 0$ is certified on every cell); (2) $E(a, -t) = E(a, t)$ (the reflection $\mathrm{diag}(1, -1)$ preserves L_ρ and conjugates P_t to P_{-t}, B2 §2.4; true for every radial V). At the window edges $t = \pm t_0$ the differentiability needs a neighbourhood of the closed window; the certified margins are strict, and the gap is closed by continuity (stated, not separately certified, as in N2).

Step 7: from the path to lattice shapes [C-human]. Since $\tau(t) = \rho + \sqrt{3}\,t + O(t^2)$ (B2 §2.5), small $t \neq 0$ give shapes $\neq \rho \bmod SL_2(\mathbb{Z})$ arbitrarily close to ρ. Hence T4 (inside) says: arbitrarily near ρ there are unit-covolume lattices with strictly smaller E_{zp}, so ρ is *not a local minimum over shapes*; no criticality or isotropy is used. For the outside side see Section 6.

4 Tables

4.1 Certified segments and root enclosures

Range of a	sign of F	records	widths	worst enclosure end	Lean
[0.425, 1.579] (inside)	< 0	71	0.001–0.02	$F \leq -0.01519$ on $[1.576, 1.579]$	`segment_neg0`
[1.584, 4] (outside)	> 0	103	0.003–0.0538	$F \geq 0.01558$ on $[1.584, 1.587]$	`segment_pos0`
[0.381, 0.383]	< 0	1	0.002	$F \leq -0.0549$	`sign_neg`

Table 2: Whole-interval certificates [C-ball] (logic [A]). Every statement is for all a in the range and all $|t| \leq 10^{-8}$. 197 records in all: 174 in the two chains, 1 interval, 22 thin points.

bracket	left record	right record	conclusion
a_-	$a = 0.3745$: $F \in [0.017732, 0.094875] > 0$	$a = 0.379$: $F \in [-0.097769, -0.023538] < 0$	zero in (0.3745, 0.379)
a_+	$a = 1.579$: $F \in [-0.042254, -0.019463] < 0$	$a = 1.583$: $F \in [0.007479, 0.030273] > 0$	zero in (1.579, 1.583)

Table 3: Root enclosures ([C-ball] for the four signs, [A] for the IVT, [C-human] for continuity). Thin points use $N = 64$.

4.2 Not certified (reported, not hidden)

The preregistered target was a full "iff" on a compact range. It is reached on the right of the window only ($[0.425, 4]$); on the left of a_- the kill rule applied (best certified sub-intervals reported). Superseded runs: wide

side	sign	thin points a with $F(a,t)$ enclosure ($\lvert t\rvert \le 10^{-8}$)
left of a_-	$F>0$	0.33: $[0.9614, 1.0893]$; 0.35: $[0.5826, 0.6914]$; 0.36: $[0.3609, 0.4639]$; 0.37: $[0.1200, 0.2207]$; 0.372: $[0.0698, 0.1702]$; 0.3745: $[0.0177, 0.0949]$
inside, near a_-	$F<0$	0.379: $[-0.0978, -0.0235]$; 0.381: $[-0.1635, -0.0635]$; 0.383: $[-0.2169, -0.1170]$; 0.39: $[-0.4074, -0.3078]$; 0.40: $[-0.6920, -0.5868]$; 0.42: $[-1.2851, -1.1723]$; 0.45: $[-2.2103, -2.0899]$; 0.5: $[-3.7455, -3.6229]$; 0.55: $[-5.1531, -5.0346]$
inside, near a_+	$F<0$	1.57: $[-0.1967, -0.0903]$; 1.576: $[-0.1215, -0.0151]$; 1.579: $[-0.0423, -0.0195]$
outside, near a_+	$F>0$	1.583: $[0.0075, 0.0303]$; 1.584: $[0.0199, 0.0427]$; 1.587: $[0.0152, 0.1216]$; 1.593: $[0.0890, 0.1956]$

Table 4: The 22 thin-point records ([C-ball]; signs [A]): 10 with $F>0$ (6 left of a_-, 4 outside) and 12 with $F<0$ (9 near a_-, 3 near a_+). With the chains this gives the 197 Lean records: 84 negative = 71 (chain) + 1 (interval $[0.381, 0.383]$) + 12 thin, and 113 positive = 103 (chain) + 10 thin. They are pointwise statements; the whole-interval statements are in the first table.

Region	status
(0.33, 0.3745)	only the thin points above; no whole interval certified (the sums grow with the lattice truncation and with R)
(0.383, 0.425)	thin points $0.39, 0.40, 0.42$ certified $F<0$; intervals of width 0.0025–0.005 on $[0.3925, 0.41]$ not certified (failed cells: $12, 9, 5, 12, 3$ in order; in earlier attempts R reached 32); $[0.41, 0.415]$, $[0.415, 0.42]$, $[0.42, 0.425]$ have enclosures $[-33.4, 31.3]$, $[-4.86, 2.54]$, $[-2.70, 0.09]$, sign undetermined
$a<0.33$	thin points $a=0.25$ and $a=0.30$ *failed*: $\det B>0$ could not be certified on 104 resp. 9 cells near Γ at refinement depth 2 (the long-wave shear mode softens towards $a\approx 0.123$); nothing is claimed
$a>4$	not attempted (P2's values at $a=6,8$ are precision-flagged)
$[a_-, a_+]$ roots	existence only; no uniqueness; the statement "$F>0$ on $(1.583, 1.584)$" is not whole-interval certified

Table 5: What failed or was not attempted. The 12 unsigned runs (7 with failed cells, 5 inconclusive) are in `data/runs/` and `data/cert_summary.json` (`uncertified`); they never entered Lean. Of the 5 inconclusive runs only 2 straddle a root ($[0.3745, 0.379]$ and $[1.579, 1.583]$); the other 3 ($[0.41, 0.415]$, $[0.415, 0.42]$, $[0.42, 0.425]$) lie where $F<0$ is expected and their enclosures are merely too wide.

a-intervals at large a (half-widths 0.013–0.036) let R grow from 4.9 to 73.8 (11 of 12 uncertified) and were re-run at width 0.025 (72 of 72 certified); only the re-runs are inputs. B2-style a-balls of relative half-width $\sim 2.5\%$ could not certify $\det B>0$ (B moves $\sim 30\%$ across the ball); this is why the affine form with lockstep tightening was built.

5 Controls

Every number above passed a positive and a negative control (LL.md rule).

- **(-) Just inside, $a=0.40$: must not certify "local minimum".** The enclosure is $F(0.40)\in[-0.692, -0.587]$, negative. `RejectInside.lean` claims `allPos [rIn] = true` for the real record: rc 1, but the failure is at elaboration (`decide` "did not reduce"/"Reduction got stuck"), so rc 1 alone does not distinguish "false" from "stuck". The negative proof is `Honest.lean`: it proves `allPos [r] = false` and `allNeg [r] = true` for the record by `decide +kernel` (rc 0, standard axioms, kernel replay rc 0) [A]; `RejectInside`/`RejectTampered` only show that the false claim cannot be proved. Also $0.379, 0.381, 0.383, 0.39$ are certified negative.
- **(-) Tampered certificate.** A real record ($a=0.372$) with the sign of q_{lo} flipped: `RejectTampered.lean`

rc 1 (again an elaboration failure); `Honest.lean` proves `tampered_not_pos` and `untampered_pos` (rc 0) [A].

- **(-) Straddling intervals certify neither sign.** $[0.3745, 0.379]$: $F \in [-0.1514, 0.1470]$, sign "none"; $[1.579, 1.583]$ ($N = 64$): $F \in [-0.0482, 0.0362]$, "none" (`data/runs/ctl_straddle*.json`) [C-ball]. The certifier does not fabricate a sign across a root, and the records bracket it exactly as T3 says.
- **(-) Planted `sorry` and planted axiom (`Controls.lean`).** It compiles with rc 0; the axiom gate shows `sorryAx` for `ctl_sorry` and `ProgZeroPointClass.ctl_cheat` for `ctl_axiom`; the honest `ctl_false` prints the standard list [A].
- **(-) LeanFlow, 197 signs and a planted false one.** The unmodified LeanFlow `lean.certify` (commit 06084ef, `decide_kernel`) on the 197 exact corner bounds: status ok, code verified, tier A (for the arithmetic statement only), all axiom reports standard. A copy with the first sign planted false: rejected, `false-statement`, failed `cert_0` [A] (`data/leanflow/`).
- **(+) Tier X cross-check, 609 checks.** Every certified enclosure contains the independent floating-point value of $F(a, 0)$ (closed form, Gauss–Legendre $N = 32$, `n1_crosscheck.py`) at a_{lo}, a_c, a_{hi} of every run: 609 checks ($= 3 \times 203$, duplicate record included), 0 not contained (`data/crosscheck_tierX.json`) [X]. Of the 210 run records, 198 carry a sign and 12 do not (7 with failed cells, 2 straddling a root, 3 inconclusive because too wide); the 12 are listed in Section 4.2.
- **(+) Structure of the engine.** The f-helper $f(x) = (1 - \cos x)/x^2$ and its derivatives: integral representation $f = \int_0^1 (1-u)\cos(ux)\,du$ (sympy residual 0) and $|f^{(k)}| \le 1/((k+1)(k+2))$; 0 of 33,000 random ball-times-sample points outside (`runs/fcheck.txt`) [X]. At $a = 1$ the first-order engine encloses E'' in $[-7.768, -7.623]$ against [X] -7.6959 and B2's independent enclosure of $U''(0)$ overlaps [C-ball].

Not done: a diagonal-path cross-check of the full engine (it supports the shear path only); B2's majorant code was not re-run. The cross-check is floating point and independent of the Arb code only in the sense of using a different evaluation of the same formula; it checks consistency, not the formula.

6 Scope of "strict local minimum over shapes": extra [C-human] assumptions

T5 is a statement along one curve through ρ. To pass to "ρ is a strict local minimum of E_{zp} over unit-covolume lattice shapes" for $a \in [1.584, 4]$ (T6) one needs, *all as written arguments*:

1. E_{zp} is of class C^2 on a neighbourhood of ρ in the shape space (the upper half plane modulo $SL_2(\mathbb{Z})$), including C^2 in the shape coordinates, not only along the shear;
2. ρ is a critical point of E_{zp} and its Hessian is isotropic. Both follow from the order-3 stabiliser of ρ (P2 §3.1, [C-human]/[X]), which acts on the tangent plane by a rotation of angle $2\pi/3$, so an invariant quadratic form is a multiple of the metric and an invariant linear form is 0; they use C^2;
3. the identification $F(a, 0) = 2 \operatorname{Lap} E_{\mathrm{zp}}(\rho)$ (B2 §2.6; it uses $|\tau'(0)|^2/y^2 = 4$ at ρ, i.e. the normalisation of the shear parameter);
4. the 2D second-derivative test: a positive definite Hessian at a critical point is a strict local minimum.

Item 4 is classical analysis and not formalised here; the Lean of P2 only contains the algebra `rot120_comm_symm` and `rot120_fixed_vec`. With these four, $F(a, 0) > 0$ gives a positive definite Hessian, and T6 follows. Without them the proven statement is T5 (strict minimum *along the shear*). Dually, T4 does not need any of them because a single descent direction suffices. Nothing in this section is used by T1–T5, T7.

7 The certifier, briefly

Everything here is [C-ball] and lives in `data/code/n1_certify2.py` (sha256 below); it is not kernel-checked.
Jets. Every quantity is a second-order forward-mode jet in (s, t', a) over the cell ball, the a-interval and the t-window, computed in lockstep with a first-order jet at the thin centre; after each operation the ball value

and gradient are replaced by mean-value forms around the centre and intersected with the naive enclosure. This tightening is essential: without it $\det B > 0$ cannot be certified at small a. Per-n Gaussian Hessians over the a-interval use Taylor forms in a around a_c. **Lattice tail.** One bound is added to every jet component of every entry: shells of width 1/4, polynomial factors at the outer radius and Gaussian factors at the inner radius (no monotonicity assumed), point count $\pi(r+1)^2$, weight bounds for all partials of $\theta^2 f(s\theta)$ up to order 2, an a-factor $1 + L + L^2$, and a geometric remainder whose ratio is *computed* and must be < 1. (B2's audit had flagged two shortcuts there: evaluation at the inner shell radius and an asserted ratio; both are fixed.) **Cost.** Small a needs many lattice terms (99 half-terms at $a = 0.37$, 147 at 0.25) and a nearly singular B near Γ; R grows fast with the a-width. Total sweep CPU: 13,685 s over the 210 run records, 13,599 s over the distinct records, the duplicate excluded (cores 4,5; other users' jobs shared core 5 for part of the time). Settings: $N = 32$ or 64, depth 1 or 2, 96 bits, $t_0 = 10^{-8}$. Cells certified: 1,284,310 (232,178 refined) over the 203 run records with ok status, one of which ($a = 1.584$) is a duplicate; over the 202 distinct records 1,276,022 (232,146 refined); 0 failed. Of the 203 ok records, 198 carry a sign and 5 are inconclusive (2 straddling a root, 3 too wide).

8 Lean formalisation and LeanFlow

Modules. `formal_cert/Programmes/ZeroPointClass/`: `Basic.lean` (imports B2's `Programmes.ZeroPointCert.Basic`, Mathlib submodules only), `Cert.lean` (generated by `n1_make_cert.py` from the run JSONs, endpoints verbatim; imports Basic), `Controls.lean`, `Honest.lean`, `RejectInside.lean`, `RejectTampered.lean`; never `import Mathlib`; no `sorry` outside `Controls.lean`. Pinned Lean 4.34.1 via `scripts/certified_numerics/lean_pinned.py`: `Basic.lean` rc 0, 0 errors, 21 theorems (374.9 s); `Cert.lean` rc 0, 0 errors, 11 theorems (21.6 s); `Controls.lean` rc 0 (with the planted `sorry`/axiom); `Honest.lean` rc 0; `RejectInside.lean` and `RejectTampered.lean` rc 1 as required. Every printed declaration of `Basic.lean` and `Cert.lean` depends on exactly {`propext`, `Classical.choice`, `Quot.sound`}; verbatim lines for `Cert.lean`:

```
'ProgZeroPointClass.sign_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPointClass.sign_neg' ... brackets_ordered ... classification ... along_path ... chain_pos0 ... segment_pos0
... segment_pos0_path ... chain_neg0 ... segment_neg0 ... segment_neg0_path   (all: [propext, Classical.choice, Quot.sound])
```

(Full list in `data/lean_cert.json`.)

What Tier A certifies. The window lemmas, the logic of signs and the IVT, the chain coverage, and the signs of the rational records. *It does not certify* that the records enclose F (`Encloses`), that F is the second derivative of E_{zp}, differentiation under the integral, the mirror symmetry, the continuity in a, or any step of Section 6.

Kernel replay (leanchecker). The recorded kernel replay of 2026-10-26 (`data/kernel_replay_2026-10-26_replay.jsonl`, from `results/kernel_replay_2026-10-26/`) contains, for this work, the line `Programmes.ZeroPointClass.Basic`: rc 0 (58.9 s), and for its import `Programmes.ZeroPointCert.Basic`: rc 0 (65.4 s). That recorded replay has no line for `Programmes.ZeroPointClass.Cert`. The independent AI audit (`AUDIT.md`) closed this gap: it rebuilt the oleans from the current sources in a private directory and ran leanchecker 4.34.1 on each module (`data/leanchecker_replay_audit.json`): `Programmes.ZeroPointClass.Cert` **rc 0** (55.5 s), `Controls` rc 0 (52.6 s), `Honest` rc 0 (49.9 s), `Basic` rc 0 (53.1 s); in a second full rebuild of the import chain also `Programmes.ZeroPoint.Basic` and `Programmes.ZeroPointCert.Basic` rc 0. `RejectInside` and `RejectTampered` fail to compile by design and are not replayed. Replay does not check axioms (`Controls` replays with rc 0 although it holds `sorryAx` and a planted axiom); the `#print axioms` audit above does. The `decide +kernel` proofs of the chains, signs and brackets are re-evaluated by the kernel in the `Cert` replay.

Hashes (sha256).

- `Basic.lean`: `dcb36833d0825c37a7a41fcca00cf66d8c3e8c841b506a76f4ae12b77754a346`
- `Cert.lean`: `fde4a58008a4615e3c8a979c5ce8b41cb72eac13529974d536cdac7e23c0401c`
- `Controls.lean`: `6335105560ed35d56abcc8e07187afb09b9885f0085ca64660a8fd857e7fa75d`

- Honest.lean: 9c261c805b57e188aedefe9297aa04c2fb54bfbb5406de1a100d8c7226d92fbb
- RejectInside.lean: 2bc76e222ea7bbccbe587d38f5b34a27bfacc3ac048193db90f5ba503a923fb2
- RejectTampered.lean: 7ae0e76e22f4fe199467477d2f3b14a1c0d99c60df28cf14f401adea35cb2ba4

LeanFlow. The unmodified tool (`/home/callensxavier_gmail_com/LeanFlow-wt-p1`, commit 06084ef) renders the 197 sign claims on the exact corner bounds into integer statements discharged by `decide`; claim sha256 `497f465a...`. It certifies the statement about the stated numbers, not their provenance.

9 Claim ledger

`ledger/build_ledger.py` files 27 claims: 8 at Tier A (`PZC-A-0001`–`0008`), 14 at C (6 [C-ball]: `C-0001`–`0006`; 8 [C-human]: `C-0007`–`0014`, including the headlines `C-0012`, `C-0013` and the conditional `C-0014`) and 5 at X (`X-0001`–`0005`); each has a content-addressed evidence blob in `ledger/evidence/`. The gate's tier discipline forces every headline to Tier C because it rests on [C-ball] and [C-human] rows; the Lean rows stay at A. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`; verbatim in `ledger/gate_output.txt`) exits with code **1**: **0 blocking findings and 8 flags** (`LEDGER_UNAUDITED_TIER_A`, one per Tier A row, because no *person* has certified that the Lean statements mean what their glosses say). The independent AI audit (`AUDIT.md`, ACCEPT_WITH_CORRECTIONS, no blocking item) is recorded in the `ai_audit` field of every Tier A row; the audit auditor re-ran the gate and obtained the same output. There is no human audit and no statement lock, so the gate's person-audit field stays null. Exit code 1 is the honest state, not a failure to be cleared by editing.

10 Limitations

- **Partial result.** Only $a \in [0.425, 4]$ is classified, plus isolated points left of a_-. The full iff for all $a > 0$ is not claimed, and P2's sign change near $a \approx 0.123$ (and the precision-flagged values at $a = 6, 8$) show that no simple iff can be expected. Roots are enclosed, not shown unique.
- **Mostly Tier C.** The enclosures ([C-ball]) are executed code, not proofs; the identity of F with $\partial_t^2 E$, differentiation under the integral, mirror symmetry, continuity in a and Section 6 are written arguments checked by no machine and no human referee.
- **Tiny window and one path.** $|t| \le 10^{-8}$, shear path only, harmonic approximation, Bravais lattices, two dimensions, Gaussian family only, $\hbar = m = 1$. The result concerns E_{zp} alone; the classical energy contributes separately and nothing follows for the total energy.
- **AI audit only.** One independent AI audit (ACCEPT_WITH_CORRECTIONS, no blocking item) exists; its corrections are applied here. No human has audited anything; no statement lock.
- **Kernel replay.** `Cert`, `Controls`, `Honest` and `Basic` replay with rc 0 under leanchecker (auditor-built oleans, Section 8); the replay is not an axiom audit.
- **Over-strong Lean hypotheses (known, sound).** In `Cert.lean` the hypotheses `hE`, `hE'` and `hsym` are quantified over every real a (including $a \le 0$, where E_{zp} is undefined) rather than over a in the record's range. This is not unsound (E and F are free variables and can be extended arbitrarily outside the range) and the proofs use only a in range, but it makes instantiation awkward and the hypotheses stronger than their gloss. The Lean files and generated certificates are unchanged in this revision; the fix would be to quantify over $\mathrm{lo} \le a \le \mathrm{hi}$ in `n1_make_cert.py`.
- **Repository guards fail for pre-existing reasons.** `python antigravity_guard.py` rc 1 and `python test_rigor_guard.py` rc 1 (hallucinated-import reports for packages absent from the venv, such as `datasets`, `peft`, `trl`, `flint`, and hollow tests in `tests/test_k3_anse_improvements.py`, `tests/test_scientific_rigor_v14.py`); neither output mentions any N1 file (`data/guard_*.txt`). `pytest tests/` was not run for this paper. "Guards pass" is not claimed.
- **Cross-check is consistency, not independence.** The 609-check comparison is floating point with the same formula (Section 5). The N1 engine was developed iteratively (earlier versions in `data/code/`);

the 19 Lean input files were all produced with one version, recorded in each run JSON by sha256 of `n1_certify2.py` and of the imported helper `n1_certify.py`.

- **python-flint** 0.6.0 is not in the project venv; re-running needs that version. One record ($a = 1.584$, `runs/sweep_ap_tight.json`) is a duplicate (114 positive records in the runs, 113 in Lean; the Lean list is de-duplicated: 197 records); counts of cells and CPU time are given with and without it.
- **Not novelty-checked.** No literature search was made for prior certified results on zero-point lattice energies; no novelty is claimed.

11 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: `9efd4f85456c55c883a4583cd83b7ecba2b7984a` (the N1 producer commit); the Lean files and data of this paper are present in the working tree. Nothing was pushed or published. Copies of the inputs are in `data/` (`data/code/` scripts, `data/runs/` the 19 input run files and their logs, `data/lean/` the six Lean files, `data/leanflow/`). Commands (cores 4,5; from `results/proposal3/2026-10-26/N1`):

```
python -I code/n1_explore.py 64 0.15 0.2 ... 4          # Tier X, runs/explore_tierX_N64.txt
python -I code/n1_fcheck.py                             # runs/fcheck.txt
python -I code/n1_certify2.py OUT.json N 1/100000000 0 TOL DEPTH A_LO:A_HI ...   # every run JSON; queues via code/n1_queue.py
python -I code/n1_make_cert.py ../../../../formal_cert/Programmes/ZeroPointClass/Cert.lean cert_summary.json 749/2000 379/1000
      1579/1000 1583/1000 <the 19 run files of cert_summary.json "inputs">
python -I code/n1_make_controls.py ../../../../formal_cert/Programmes/ZeroPointClass 2/5 93/250 runs/sweep_pts_extra.json runs
     /sweep_left_pts.json
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPointClass/Basic.lean --olean Programmes.
     ZeroPointClass.Basic
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPointClass/Cert.lean --olean Programmes.
     ZeroPointClass.Cert   # then Controls, Honest, RejectInside, RejectTampered
python -I code/n1_leanflow.py leanflow <run files>      # LeanFlow: 197 verified, planted false rejected
python -I code/n1_crosscheck.py crosscheck_tierX.json <run files>   # 609 checks, 0 misses
python -I ledger/build_ledger.py ; python -I ledger/run_gate.py     # in papers/p3_zero_point_classification; gate: exit 1 = 0
      blocking, 8 flags
```

Certifier hashes (sha256, recorded in every run JSON):

- `n1_certify2.py`: `e4f11b18e2b42b0775344272dde009ad95c232bb298ecb6922844447496a2452`
- `n1_certify.py` (imported helpers): `f7ac95ddcf7acf1fb3e905120242511ed8a7b22197fcca07a2f10980994e74b5`

References

[1] X. Callens, *Does the triangular lattice minimise the harmonic zero-point energy? A numerical counterexample for completely monotone potentials, with kernel-checked curvature identities*, Zenodo, doi:10.5281/zenodo.23653833 (2026). Own preprint (P2), source of the conjecture QU-1 and of the [X] roots and the sign change near 0.123; not peer reviewed.

[2] X. Callens, *The triangular lattice is not a local minimum of the harmonic zero-point energy for* $\exp(-\pi r^2)$*: a certified linear-response majorant along a symmetric shear*, Zenodo, doi:10.5281/zenodo.23655392 (2026). Own preprint (B2); its Lean module `Programmes.ZeroPointCert.Basic` is imported; not peer reviewed.

[3] X. Callens, *The triangular lattice is not a local minimum of the harmonic zero-point energy for planar Yukawa potentials* $\kappa = 1, 2 \ldots$, Zenodo, doi:10.5281/zenodo.23663551 (2026). Own preprint (N2), model for the format, ledger and audit-status conventions; not peer reviewed.

[4] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, arXiv:1611.02831 (2016); cited for the instrument only.

[5] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019), doi:10.1145/3372885.3373824 (CPP 2020); cited for the library only.